CONCENTRATED PRESSURE: RESOURCE CONCENTRATION AND ETHICAL AGENT OUTCOMES IN THE SUGARSCAPE DIGITAL TERRARIUM

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SUBMITTED TO THE FACULTY OF THE GRADUATE SCHOOL
UNIVERSITY OF THE PHILIPPINES LOS BAÑOS
IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE
DEGREE OF

BACHELOR OF SCIENCE
(Computer Science)

MAY 2026

The thesis hereto attached entitled CONCENTRATED PRESSURE: RESOURCE CONCENTRATION AND ETHICAL AGENT OUTCOMES IN THE SUGARSCAPE DIGITAL TERRARIUM, prepared and submitted by JOMAR MONREAL, in partial fulfillment of the requirements for the degree of BACHELOR OF SCIENCE (Computer Science) is hereby accepted.

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JOMAR MONREAL

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ABSTRACT

JOMAR MONREAL, University of the Philippines Los Baños, MAY 2026. **CONCENTRATED PRESSURE: RESOURCE CONCENTRATION AND ETHICAL AGENT OUTCOMES IN THE SUGARSCAPE DIGITAL TERRARIUM** Major Professor: PROF. EDSGER W. DIJKSTRA

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INTRODUCTION

Background of the Study

Ethical decision-making is a central challenge in designing modern computational systems. As artificial agents become more capable of autonomous action, researchers need ways to test how ethical decision rules shape both individual behavior and collective outcomes. Yet many ethically significant situations cannot be tested in real life, since doing so may cause harm, violate consent, or create unacceptable risks. Agent-based modeling (ABM) offers a practical alternative: it lets researchers simulate societies of interacting agents and observe how individual choices generate broader social outcomes. ABM enables virtual ethical experiments that would be impermissible in real settings and supports inquiry where direct data collection would violate ethical constraints (Lasquety-Reyes, 2018; Slater et al., 2006; Groff et al., 2018).

The Digital Terrarium developed by Kremer-Herman et al. (2024) extends the standard Sugarscape agent-based simulation with a socio-ethical decision layer. This lets agents follow ethical models rather than purely greedy behavior focused on maximizing resources. The framework implements three ethical theories: Bentham's hedonic act utilitarianism, ethical altruism, and ethical egoism. It also allows researchers to compare how these ethical orientations shape survival, resource distribution, and societal stability. Kremer-Herman et al. (2024) reported that utilitarian societies consistently outperformed altruistic and egoistic ones, with larger populations, greater accumulated wealth, and longer survival (p. 195607). These findings mark a key step in computational ethical modeling and show that the chosen ethical decision model has measurable societal effects.

The way resources are distributed in the simulation environment also affects how agents behave and what outcomes emerge at the societal level. In agent-based simulation, the environment provides the infrastructure and social drivers that shape agent behaviors and policy outcomes (Tolk et al., 2022, pp. 90, 92). Hawick (2014) demonstrated this directly in Sugarscape by comparing a twin-peak resource configuration against a uniformly distributed flat field. Under the twin-peak setup, two sugar mounds were placed at opposing corners of the grid using a sinusoidal distribution, creating concentrated zones of high resource availability surrounded by resource-poor plains. Hawick found that this peak configuration dramatically affects the inequalities in agent opportunity and population survival times. Specifically, agents located near resource peaks become super-rich while those on the flat plains starve to death, and the system as a whole lasts longer under concentrated, unequal resource conditions than under evenly distributed ones (pp. 2-3, 6). Despite this, Hawick's investigation was limited to greedy agents with no ethical model. TFagan et al. (2022) demonstrated in a theoretical ecology framework that the number of resource patches fundamentally shapes consumer outcomes. In a single-patch landscape, resource matching success was lower and more variable than in a two-patch landscape, because resources concentrated in one location are harder to locate and exploit efficiently (p. 276). This indicates that reducing the number of resource peaks is a meaningful manipulation that amplifies competitive pressure on agents. In turn, Kremer-Herman et al. (2024) tested their ethical decision models exclusively under a symmetric four-peak resource configuration: two sugar peaks and two spice peaks distributed across opposing quadrants. It creates a balanced landscape in which resources remain accessible across most of the simulation environment.

No study has examined how ethical decision models perform when resource access is geographically concentrated rather than distributed. Kremer-Herman et al. (2024) found that non-utilitarian societies frequently went extinct because they were not suited to survive in a hostile environment (p. 195600). However, the resource configuration of that environment was kept the same throughout their study. It remains unknown whether their findings hold when the environment itself is made more extreme.

This study addresses that gap by examining the societal outcomes of ethical decision models in the Sugarscape Digital Terrarium under a single-peak resource configuration, in which all resources are concentrated in one location, leaving the remainder of the landscape resource-poor. By reducing the resource landscape from four peaks to one, this study creates the condition of genuine resource scarcity that Hawick (2014) identified as producing the sharpest inequalities in agent opportunity. The study compares the performance of utilitarian, altruistic, and egoistic decision models under this concentrated environment and evaluates whether the comparative advantage of utilitarianism established by Kremer-Herman et al. (2024) under a balanced resource layout is preserved, amplified, or diminished when agents must compete under extreme resource concentration. In doing so, this study contributes to the growing field of computational ethics by examining the sensitivity of ethical model outcomes to environmental conditions, and offers evidence on whether resource geography is a variable that researchers must control when drawing conclusions from ethical agent simulations.

Statement of the Problem

Kremer-Herman et al. (2024) showed that ethical decision models produce different societal outcomes in agent-based simulation, but their experiments were conducted under a single, fixed resource environment. It is not known whether those results hold when resource access is concentrated rather than distributed.

Objectives of the Study

The general objective of this study is to examine the effects of resource concentration on the societal outcomes of ethical decision models in the Sugarscape Digital Terrarium simulation, and to assess how the composition of mixed ethical societies interacts with resource geography. Specifically, this study aims to:

1. compare the societal outcomes of utilitarian, altruistic, and egoistic decision models

under the distributed resource configuration ($d = 14$) as an experimental baseline;

2. compare the societal outcomes of utilitarian, altruistic, and egoistic decision models across a gradient of peak distances ($d = 14$, $d = 7$, $d = 0$);

3. determine whether the performance advantage of the utilitarian decision model at $d = 14$ is preserved, amplified, or diminished as resource peaks converge toward the center; and

4. examine how varying the proportion of utilitarian agents in mixed Egoist-Bentham societies affects societal outcomes under both the distributed ($d = 14$) and concentrated ($d = 0$) configurations.

REVIEW OF LITERATURE

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METHODOLOGY

Research Design

This study used a computational experimental research design using agent-based simulation. The study examined how resource concentration affects the societal outcomes of ethical decision models in the Sugarscape Digital Terrarium, and how ethical model composition in mixed societies interacts with resource geography. The research design was experimental because the study compared different decision-making conditions and resource configurations within the same simulation environment.

The study was conducted in two phases. Phase 1 manipulated two independent variables: resource configuration and ethical decision model. Resource configuration was operationalized as the radial distance d of the four resource peaks from the center of the grid. All configurations maintained four peaks: two sugar peaks and two spice peaks placed symmetrically at a common distance d from the center in opposing diagonal quadrants. At $d = 14$, the peaks occupied the corner quadrants of the grid, replicating the layout of Kremer-Herman et al. (2024) and representing the most distributed resource environment. At $d = 7$, the peaks were drawn inward toward the center, representing an intermediate level of concentration. At $d = 0$, all four peaks converged on the central cell, producing the most concentrated resource environment. This convergence at $d = 0$ is mathematically equivalent to a single co-located resource peak, as confirmed by the simulation source: the environment assigns each cell the maximum resource capacity across all overlapping peaks rather than summing them. The use of a single distance parameter d creates a principled and continuous concentration gradient, isolating peak separation as the manipulated variable while holding the number of peaks constant. The ethical decision model had three levels: utilitarian, altruistic, and egoistic, corresponding to the decision models implemented by Kremer-Herman et al. (2024). These two variables produced a $3{\times}3$ experimental design with nine conditions in total.

Table 1. Phase 1 Experimental Conditions

Condition	Resource Configuration	Decision Model
1	$d = 14$ (distributed, control)	Utilitarian
2	$d = 14$ (distributed, control)	Altruistic
3	$d = 14$ (distributed, control)	Egoistic
4	$d = 7$ (intermediate)	Utilitarian
5	$d = 7$ (intermediate)	Altruistic
6	$d = 7$ (intermediate)	Egoistic
7	$d = 0$ (concentrated, treatment)	Utilitarian
8	$d = 0$ (concentrated, treatment)	Altruistic
9	$d = 0$ (concentrated, treatment)	Egoistic

Conditions 1 through 3 replicated the original Digital Terrarium experiments of Kremer-Herman et al. (2024) at $d = 14$ and served as the experimental baseline. Conditions 4 through 6 introduced the intermediate $d = 7$ configuration. Conditions 7 through 9 used the fully concentrated $d = 0$ layout and constituted the most novel experimental contribution of this study. Together, the three distance levels form a gradient of decreasing resource accessibility, allowing the study to examine how progressive peak convergence affects the societal outcomes of each ethical decision model.

Phase 2 extended the design to include heterogeneous societies, following the mixed-agent investigation of Kremer-Herman et al. (2024). The proportion of utilitarian (Bentham) agents was varied at 20%, 40%, 60%, and 80% within mixed Egoist-Bentham societies under the $d = 14$ and $d = 0$ configurations. The pure 0% (fully egoistic) and 100% (fully utilitarian) endpoints were already covered by Phase 1. This produced four novel proportion levels per configuration, adding eight heterogeneous conditions in total.

The dependent variables measured were population size over time, population end state, mean agent wealth, total societal wealth, Gini coefficient of wealth distribution, and mean life expectancy. All other simulation parameters were held constant across all

Table 2. Phase 2 Heterogeneous Conditions (Egoist-Bentham Sweep)

Conditions	**Resource Configuration**	**% Utilitarian Agents**
10–13	$d = 14$ (distributed, control)	20%, 40%, 60%, 80%
14–17	$d = 0$ (concentrated, treatment)	20%, 40%, 60%, 80%

conditions, including grid size at 50 by 50 cells, starting agent population at 250, and all remaining Digital Terrarium default settings. Each condition was run across 30 independent random seeds to account for stochastic variation. Phase 1 produced 270 simulation runs and Phase 2 produced 240 runs, for a total of 510 simulation runs across the study. Results across seeds were averaged to obtain stable estimates for each condition.

Figure 1. Test Caption Figure

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RESULTS AND DISCUSSION

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SUMMARY AND CONCLUSION

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